Exercise on Decreasing Numbers

- Assume that in any decimal number the leftmost digit cannot be 0. How many 6-digit numbers exist such that their digits from left to right (i.e., no two digits are identical)
- **(a)** Are strictly decreasing (e.g., 864 320). Hint: Any six different digits give you a unique number with strictly decreasing digits.

  By choosing any 6 digits from $\{0, 1, 2, \cdots, 9\}$ we can form a unique 6 digit number that its digits are strictly decreasing from left to right. The answer is C(10,6)
- **(b)** Are strictly increasing (e.g., 136 789).

  Similar to part (a), but we can not pick 0 because then it has to be the leftmost digit. So, the answer is C(9,6)